Jianhao Zeng

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Education

Tianjin University — Tianjin, China
M.Eng. in Electronic and Information Engineering — 2021/09 – 2024/06
Advisor: Prof. Dan Song

Tianjin University — Tianjin, China
B.Eng. in Mechanical Design & Manufacturing and Their Automation — 2017/09 – 2021/06

Experiences

Machine Learning Department, AMAP, Alibaba Group — Beijing, China
Algorithm Engineer — 2025/06 – Current
Mentor: Dr. Lei Sun, Dr. Yancheng Bai and Mr. Xiangxiang Chu
Research topics: image/video generation, world models, and autonomous agents

MAPLE Lab, Westlake University — Hangzhou, China
Research Assistant — 2024/06 – 2025/01
Advisor: Dr. Liyuan Ma, Dr. Zhiyang Chen and Prof. Guojun Qi (Fellow of IEEE, IAPR and AAIA)
Research topics: video generation and diffusion models

Institute of Television and Image Information, Tianjin University — Tianjin, China
Graduate Student — 2021/09 – 2024/06
Advisor: Prof. Dan Song and Prof. Anan Liu (Distinguished Young Scholars)
Research topics: image generation and controllable generation

Publications

1. **Elucidating the SNR-t Bias of Diffusion Probabilistic Models**
   Meng Yu, Lei Sun, Jianhao Zeng, Xiangxiang Chu, Kun Zhang#
   CVPR 2026
2. **Eevee: Towards Close-up High-resolution Video-based Virtual Try-on**
   Jianhao Zeng*, Yancheng Bai*, Ruidong Chen, Xuanpu Zhang, Lei Sun, Dongyang Jin, Ryan Xu, Nannan Zhang#, Dan Song, Xiangxiang Chu
   CVPR 2026 (Findings)
3. **Layer-wise Instance Binding for Regional and Occlusion Control in Text-to-Image Diffusion Transformers**
   Ruidong Chen, Yancheng Bai, Xuanpu Zhang, Jianhao Zeng, Lanjun Wang, Dan Song, Lei Sun, Xiangxiang Chu, Anan Liu#
   CVPR 2026
4. **Semantic Context Matters: Improving Conditioning for Autoregressive Models**
   Dongyang Jin*, Ryan Xu*, Jianhao Zeng, Rui Lan, Yancheng Bai#, Lei Sun#, Xiangxiang Chu
   CVPR 2026
5. **Group Relative Attention Guidance for Image Editing**
   Xuanpu Zhang*, Xuesong Niu*, Ruidong Chen, Dan Song, Jianhao Zeng, Penghui Du, Haoxiang Cao, Kai Wu#, Anan Liu#
   CVPR 2026 (Findings)
6. **MEF-GD: Multimodal Enhancement and Fusion Network for Garment Designer**
   Dan Song, Juan Zhou, Jianhao Zeng, Hongshuo Tian, Bolun Zhen, Rongbao Kang, Anan Liu#
   TCSVT
7. **Robust-MVTON: Learning Cross-Pose Feature Alignment and Fusion for Robust Multi-View Virtual Try-On**

Nannan Zhang*, Yijiang Li*, Dong Du#, Zheng Chong, Zhengwentai Sun, Jianhao Zeng, Yusheng Dai, Zhenyu Xie, Hairui Zhu, Xiaoguang Han#
CVPR 2025

8. **BooW-VTON: Boosting In-the-Wild Virtual Try-On via Mask-Free Pseudo Data Training**
   Xuanpu Zhang, Dan Song#, Pengxin Zhan, Tianyu Chang, Jianhao Zeng, Qingguo Chen, Weihua Luo, Anan Liu#
   CVPR 2025
9. **Better Fit: Accommodate Variations in Clothing Types for Virtual Try-on**
   Dan Song, Xuanpu Zhang, Jianhao Zeng, Pengxin Zhan, Qingguo Chen, Weihua Luo, Anan Liu#
   TCSVT
10. **CAT-DM: Controllable Accelerated Virtual Try-on with Diffusion Model**
    Jianhao Zeng, Dan Song#, Weizhi Nie, Hongshuo Tian, Tongtong Wang, Anan Liu#
    CVPR 2024
11. **Fashion Customization: Image Generation Based on Editing Clue**
    Dan Song, Jianhao Zeng, Min Liu, Xuanya Li, Anan Liu#
    TCSVT

Services & Activities

- **Reviewer**: ACM MM (2024), ICLR (2025, 2026), NeurIPS (2025, 2026), CVPR (2026), ECCV (2026), TCSVT
- **Teaching Assistant**: Digital Logic Circuit, Tianjin University
- **Translation**: Physically Based Rendering: From Theory To Implementation, fourth edition
- **Patent**: A Fashion Image Editing Method and Device Based on Self-Attention Mechanism (CN115082295B)

Awards

- CVPR Registration and Travel Support — 2024
- Excellent Master's Degree Thesis of Tianjin University (**Top 5%**) — 2024
- Tianjin University Academic Scholarship — 2021, 2022, 2023

Competitions

- **Top 6.9%** in Jiangsu Meteorological AI Algorithm Challenge — 2022/06
- **First Prize** in Tianjin University Undergraduate Physicists Tournament (TJUPT) — 2019/08
- **Second Prize** in National College Students Mathematical Competition — 2018/10
- **Third Prize** in Tianjin College Student Mathematics Competition — 2018/05

SKills

- **Programming Languages** — C, C++, Python, HTML, CSS, JavaScript
- **Frameworks** — PyTorch, PyTorch Lightning, Accelerate
- **Tools** — Linux, Git, LaTeX, Typst
- **Human Languages** — Mandarin, English (TOEFL iBT: 94)